

Chemical equilibrium and routes to K

After midterm · Session 5 of 6 · 180 minutes

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The question for today

Why can composition change while the equilibrium constant stays the same?

Before class: 3-page reading · 15–20 minutes

Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion, exit ticket and feedback	30

The 70-minute investigation is student work. Final block: discussion 20 min, exit ticket 5 min, feedback 3 min, closing 2 min.

Start with extent and driving force

$$n_i = n_{i,0} + \nu_i \xi, \quad n_i \geq 0$$

$$\left(\frac{dG}{d\xi} \right)_{T,P} = \Delta_r G = \Delta_r G^\circ + RT \ln Q$$

Write the feasible extent interval first. A negative derivative favors increasing extent; a positive derivative favors decreasing extent.

An interior equilibrium has zero derivative. Boundary minima need one-sided checks.

The equilibrium condition gives **K**

$$\Delta_r G = 0 \Rightarrow Q = K, \quad \ln K = -\frac{\Delta_r G^\circ}{RT}$$

Obtain standard Gibbs energy from direct data, formation properties, or H–TS at the evaluation temperature.

K depends on T and the reaction/standard-state convention. Q describes the current composition and pressure.

Additional K(T) routes are in the appendix and lab.

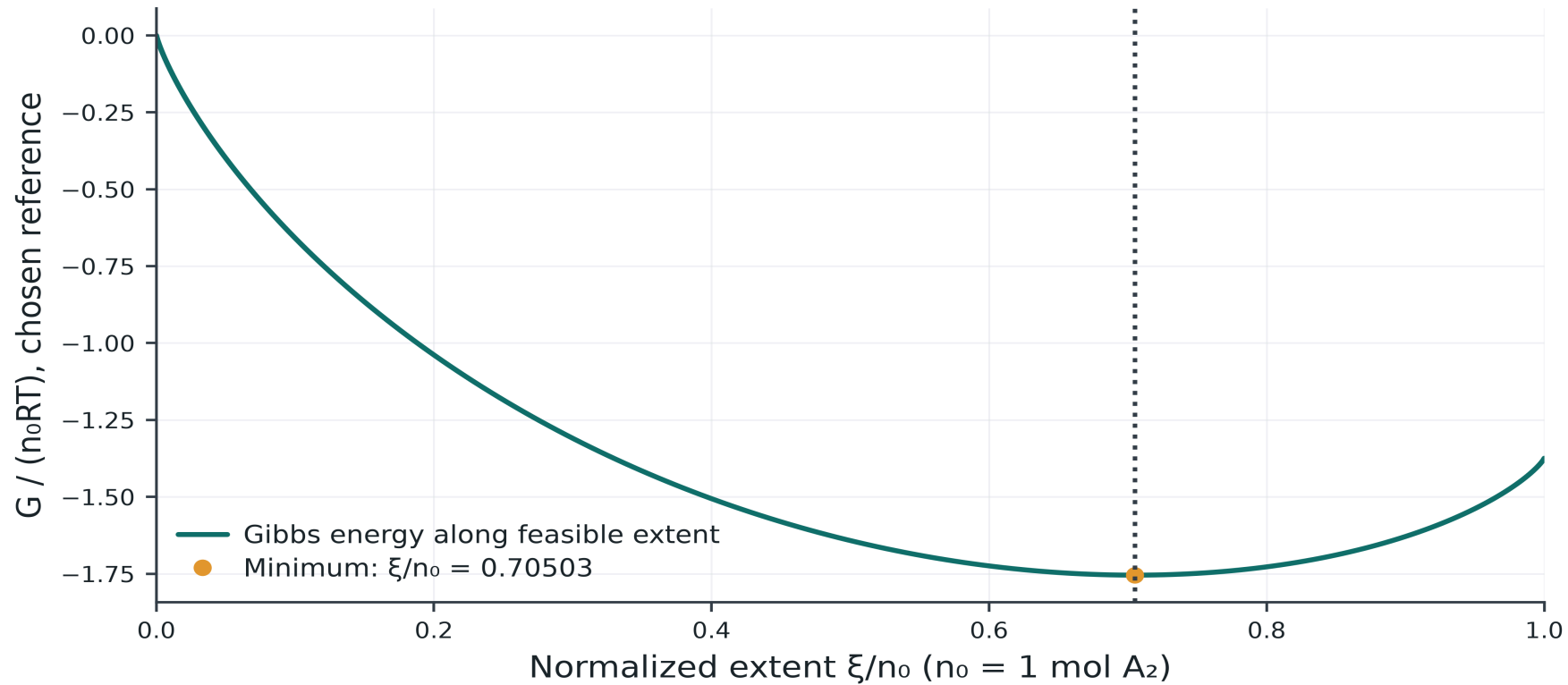
A reference-state check

Lab 12 synthetic $A_2 \rightleftharpoons 2A$ at 350 K uses $\Delta_r G^\circ = -4.000$ kJ/mol.

$$\ln K = \frac{4000}{R(350)} = 1.37454, \quad K \approx 3.95326$$

With $\Delta_r H^\circ = 50.000$ kJ/mol and $\Delta_r S^\circ = 54000/350$ J mol⁻¹ K⁻¹, the H-TS route gives the same value.

Check the minimum and the reaction quotient



Synthetic dissociation at 350 K and standard pressure. Initially one mole of reactant.

At $\xi \approx 0.70503$ mol, $Q \approx 3.95326 = K$. Check the atom balance.

One demonstration • 20 minutes

Open the core lab

Use the synthetic Lab 12 case. Write $n_{A_2}=1-\bar{\xi}$, $n_A=2\bar{\xi}$ and $0\leq\bar{\xi}\leq 1$ mol first. Compare the supplied Gibbs-energy and H-TS routes, then change only pressure and inspect K and extent.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 12, compare the supplied Gibbs-energy and H–TS routes at the reference temperature. Change pressure at fixed T and compare K with equilibrium extent. Check one balance.

Discussion • 20 minutes

Why can composition change while the equilibrium constant stays the same?

Compare results and discuss unexpected behavior. Use one numerical check to support your explanation.

After class · 30–40 minutes

In Lab 12, compare the supplied Gibbs-energy and H–TS routes at the reference temperature. Change pressure at fixed T and compare K with equilibrium extent. Check one balance.

Assessment 5 and independent checkpoint

Retain one study JSON, one comparison and a short explanation.

Optional: Session appendix.

Learning path · Offline package

Exit ticket + feedback · 8 minutes

5 min · Explain and check

Answer the session question in the form. Give one result or independent check from today's lab, with units.

3 min · Help improve the next class

Rate the pace, report lab access, and suggest one improvement.

Thai or English. No name, ID or email requested.

Wait for “**Response received.**” We will use the final 2 minutes to close the class.



Scan to submit Session 5 exit ticket and feedback.

[Open Session 5 form](#)